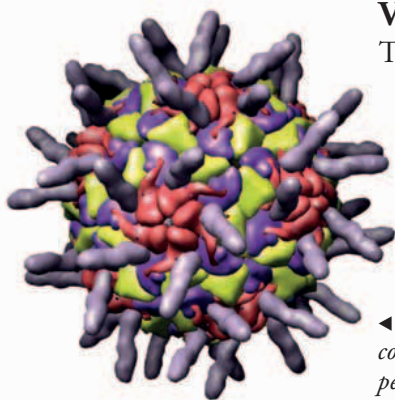


Germs and disease

Every time you breathe in or touch something, your body picks up tiny organisms that are too small to see—microorganisms. Most microorganisms do no harm, but some try to invade your body and multiply inside you, which can make you sick. These harmful microorganisms are called germs. Germs affect the body in many ways, some causing nothing more than a sneeze and others causing dangerous diseases.



VIRUSES

The tiniest germs of all are viruses. A virus is little more than a packet of genes inside a protective coat. To reproduce, it must invade one of the cells in your body, which it then hijacks and turns into a virus-making factory. Such invasions can result in colds, flu, warts, and measles.

◀ **COLD VIRUS** *The virus that causes the common cold is called a rhinovirus. It spreads from person to person in droplets that are sneezed or coughed out.*

BACTERIA

Much bigger than viruses, bacteria are single-celled organisms. Many are harmless or even helpful to your body, but some cause diseases, including food poisoning, typhoid, and plague. Invading bacteria trigger disease by releasing poisons called toxins.



▲ **BACTERIA CELLS** *These rod-shaped bacteria are called E. coli and live inside your intestines. Some types of E. coli cause diarrhea.*

WORMS

Tapeworms and roundworms are types of worms that can live inside the human body. Tapeworms live in the intestines and steal nutrients from the food you eat. They can grow to 33 ft (10 m) in length—as long as a bus.

◀ **TAPEWORM** *The head of a tapeworm is equipped with suckers and hooks, which attach it firmly to the intestine wall.*

